

Combining Equations (9.4), (9.5), and (9.9) gives the alternate form for the one-step *a priori* Kalman filter, which can be summarized as follows:

$$\begin{aligned} L_k &= F_k P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\ P_{k+1}^- &= F_k P_k^- (F_k - L_k H_k)^T + Q_k \\ \hat{x}_{k+1}^- &= F_k \hat{x}_k^- + L_k (y_k - H_k \hat{x}_k^-) \end{aligned} \quad (9.10)$$

where L_k is the redefined Kalman gain. This form of the filter obtains only *a priori* state estimates and covariances. Note that the Kalman gain, L_k , for this form of the filter is not the same as the Kalman gain, K_k , for the form of the filter that we derived in Section 5.1. However, the two forms result in identical state estimates and estimation-error covariances.

9.2 FIXED-POINT SMOOTHING

The objective in fixed-point smoothing is to obtain *a priori* state estimates of x_j at times $j+1, j+2, \dots, k, k+1, \dots$. We will use the notation $\hat{x}_{j,k}$ to refer to the estimate of x_j that is obtained by using all of the measurements up to and including time $(k-1)$. That is, $\hat{x}_{j,k}$ can be thought of as the *a priori* estimate of x_j at time k :

$$\hat{x}_{j,k} = E(x_j | y_1, \dots, y_{k-1}) \quad k \geq j \quad (9.11)$$

With this definition we see that

$$\begin{aligned} \hat{x}_{j,j} &= E(x_j | y_1, \dots, y_{j-1}) \\ &= \hat{x}_j^- \end{aligned} \quad (9.12)$$

In other words, $\hat{x}_{j,j}$ is just the normal *a priori* state estimate at time j that we derived in Section 5.1. We also see that

$$\begin{aligned} \hat{x}_{j,j+1} &= E(x_j | y_1, \dots, y_j) \\ &= \hat{x}_j^+ \end{aligned} \quad (9.13)$$

In other words, $\hat{x}_{j,j+1}$ is just the normal *a posteriori* state estimate at time j that we derived in Section 5.1. The question addressed by fixed-point smoothing is as follows: When we get the next measurement at time $(j+1)$, how can we incorporate that information to obtain an improved estimate (along with its covariance) for the state at time j ? Furthermore, when we get additional measurements at times $(j+2), (j+3)$, etc., how can we incorporate that information to obtain an improved estimate (along with its covariance) for the state at time j ?

In order to derive the fixed-point smoother, we will define a new state variable x' . This new state variable will be initialized as $x'_j = x_j$, and will have the dynamics $x'_{k+1} = x'_k$ ($k = j, j+1, \dots$). With this definition, we see that $x'_k = x_j$ for all $k \geq j$. So if we can use the standard Kalman filter to find the *a priori* estimate of x'_k then we will, by definition, have a smoothed estimate of x_j given measurements up to and including time $(k-1)$. In other words, the *a priori* estimate of x'_k will be equal to $\hat{x}_{j,k}$. This idea is depicted in Figure 9.4.

Our original system is given as

$$\begin{aligned} x_k &= F_{k-1} x_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \end{aligned} \quad (9.14)$$

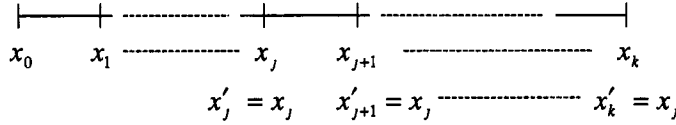


Figure 9.4 This illustrates the idea that is used to obtain the fixed-point smoother. A fictitious state variable x' is initialized as $x'_j = x_j$ and from that point on has an identity state transition matrix. The *a priori* estimate of x'_k is then equal to $\hat{x}_{j,k}$.

Augmenting the dynamics of our newly defined state x' to the original system results in the following:

$$\begin{aligned} \begin{bmatrix} x_k \\ x'_k \end{bmatrix} &= \begin{bmatrix} F_{k-1} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} x_{k-1} \\ x'_{k-1} \end{bmatrix} + \begin{bmatrix} I \\ 0 \end{bmatrix} w_{k-1} \\ y_k &= \begin{bmatrix} H_k & 0 \end{bmatrix} \begin{bmatrix} x_k \\ x'_k \end{bmatrix} + v_k \end{aligned} \quad (9.15)$$

If we use a standard Kalman filter to obtain an *a priori* estimate of the augmented state, the covariance of the estimation error can be written as

$$E \left[\begin{pmatrix} x_k - \hat{x}_k^- \\ x_j - \hat{x}_{j,k} \end{pmatrix} \begin{pmatrix} (x_k - \hat{x}_k^-)^T & (x_j - \hat{x}_{j,k})^T \end{pmatrix} \right] = \begin{bmatrix} P_k & \Sigma_k^T \\ \Sigma_k & \Pi_k \end{bmatrix} \quad (9.16)$$

The covariance P_k above is the normal *a priori* covariance of the estimate of x_k . We have dropped the minus superscript for ease of notation, and we will also feel free to drop the minus superscript on all other quantities in this section with the understanding that all estimates and covariances are *a priori*. The Σ_k and Π_k matrices are defined by the above equation. Note that at time $k = j$, Σ_k and Π_k are given as

$$\begin{aligned} \Sigma_j &= E[(x_j - \hat{x}_{j,j})(x_j - \hat{x}_j^-)^T] \\ &= E[(x_j - \hat{x}_j^-)(x_j - \hat{x}_j^-)^T] \\ &= P_j \\ \Pi_j &= E[(x_j - \hat{x}_{j,j})(x_j - \hat{x}_{j,j})^T] \\ &= E[(x_j - \hat{x}_j^-)(x_j - \hat{x}_j^-)^T] \\ &= P_j \end{aligned} \quad (9.17)$$

The Kalman filter summarized in Equation (9.10) can be written for the augmented system as follows:

$$\begin{bmatrix} \hat{x}_{k+1}^- \\ \hat{x}_{j,k+1}^- \end{bmatrix} = \begin{bmatrix} F_{k-1} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}_k^- \\ \hat{x}_{j,k}^- \end{bmatrix} + \begin{bmatrix} L_k \\ \lambda_k \end{bmatrix} \left(y_k - \begin{bmatrix} H_k & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_k^- \\ \hat{x}_{j,k}^- \end{bmatrix} \right) \quad (9.18)$$

where L_k is the normal Kalman filter gain given in Equation (9.10), and λ_k is the additional part of the Kalman gain, which will be determined later in this section. Writing Equation (9.18) as two separate equations gives

$$\begin{aligned} \hat{x}_{k+1}^- &= F_{k-1} \hat{x}_k^- + L_k (y_k - H_k \hat{x}_k^-) \\ \hat{x}_{j,k+1}^- &= \hat{x}_{j,k}^- + \lambda_k (y_k - H_k \hat{x}_k^-) \end{aligned} \quad (9.19)$$

The Kalman gain can be written from Equation (9.10) as follows:

$$\begin{aligned} \begin{bmatrix} L_k \\ \lambda_k \end{bmatrix} &= \begin{bmatrix} F_{k-1} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} P_k & \Sigma_k^T \\ \Sigma_k & \Pi_k \end{bmatrix} \begin{bmatrix} H_k^T \\ 0 \end{bmatrix} \times \\ &\quad \left(\begin{bmatrix} H_k & 0 \end{bmatrix} \begin{bmatrix} P_k & \Sigma_k^T \\ \Sigma_k & \Pi_k \end{bmatrix} \begin{bmatrix} H_k^T \\ 0 \end{bmatrix} + R_k \right)^{-1} \\ &= \begin{bmatrix} F_{k-1} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} P_k & \Sigma_k^T \\ \Sigma_k & \Pi_k \end{bmatrix} \begin{bmatrix} H_k^T \\ 0 \end{bmatrix} (H_k P_k H_k^T + R_k)^{-1} \end{aligned} \quad (9.20)$$

Writing this equation as two separate equations gives

$$\begin{aligned} L_k &= F_k P_k H_k^T (H_k P_k H_k^T + R_k)^{-1} \\ \lambda_k &= \Sigma_k H_k^T (H_k P_k H_k^T + R_k)^{-1} \end{aligned} \quad (9.21)$$

The Kalman filter estimation-error covariance-update equation can be written from Equation (9.10) as follows:

$$\begin{aligned} \begin{bmatrix} P_{k+1} & \Sigma_{k+1}^T \\ \Sigma_{k+1} & \Pi_{k+1} \end{bmatrix} &= \begin{bmatrix} F_k & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} P_k & \Sigma_k^T \\ \Sigma_k & \Pi_k \end{bmatrix} \times \\ &\quad \left(\begin{bmatrix} F_k^T & 0 \\ 0 & I \end{bmatrix} - \begin{bmatrix} H_k^T \\ 0 \end{bmatrix} \begin{bmatrix} L_k^T & \lambda_k^T \end{bmatrix} \right) + \begin{bmatrix} Q_k & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} F_k P_k F_k^T - F_k P_k H_k^T L_k^T & -F_k P_k H_k^T \lambda_k^T + F_k \Sigma_k^T \\ \Sigma_k F_k^T - \Sigma_k H_k^T L_k^T & -\Sigma_k H_k^T \lambda_k^T + \Pi_k \end{bmatrix} + \\ &\quad \begin{bmatrix} Q_k & 0 \\ 0 & 0 \end{bmatrix} \end{aligned} \quad (9.22)$$

Writing this equation as three separate equations gives

$$\begin{aligned} P_{k+1} &= F_k P_k (F_k - L_k H_k)^T + Q_k \\ \Pi_{k+1} &= \Pi_k - \Sigma_k H_k^T \lambda_k^T \\ \Sigma_{k+1}^T &= -F_k P_k H_k^T \lambda_k^T + F_k \Sigma_k^T \\ \Sigma_{k+1} &= \Sigma_k (F_k - L_k H_k)^T \end{aligned} \quad (9.23)$$

It is not immediately apparent from the above expressions that Σ_{k+1}^T is really the transpose of Σ_{k+1} , but the equality can be established by substituting for P_k and L_k .

Equations (9.19) – (9.23) completely define the fixed-point smoother. The fixed-point smoother, which is used for obtaining $\hat{x}_{j,k} = E(x_j | y_1, \dots, y_{k-1})$ for $k \geq j$, can be summarized as follows.

The fixed-point smoother

1. Run the standard Kalman filter up until time j , at which point we have $\hat{x}_j^-$ and P_j^- . In the algorithm below, we omit the minus superscript on P_j^- for ease of notation.
2. Initialize the filter as follows:

$$\begin{aligned} \Sigma_j &= P_j \\ \Pi_j &= P_j \\ \hat{x}_{j,j} &= \hat{x}_j^- \end{aligned} \quad (9.24)$$

3. For $k = j, j + 1, \dots$, perform the following:

$$\begin{aligned}
 L_k &= F_k P_k H_k^T (H_k P_k H_k^T + R_k)^{-1} \\
 \lambda_k &= \Sigma_k H_k^T (H_k P_k H_k^T + R_k)^{-1} \\
 \hat{x}_{j,k+1} &= \hat{x}_{j,k} + \lambda_k (y_k - H_k \hat{x}_k^-) \\
 \hat{x}_{k+1}^- &= F_k \hat{x}_k^- + L_k (y_k - H_k \hat{x}_k^-) \\
 P_{k+1} &= F_k P_k (F_k - L_k H_k)^T + Q_k \\
 \Pi_{k+1} &= \Pi_k - \Sigma_k H_k^T \lambda_k^T \\
 \Sigma_{k+1} &= \Sigma_k (F_k - L_k H_k)^T
 \end{aligned} \tag{9.25}$$

As we recall from Equation (9.16), P_k is the *a priori* covariance of the standard Kalman filter estimate, Π_k is the covariance of the smoothed estimate of x_j at time k , and Σ_k is the cross covariance between the two.

9.2.1 Estimation improvement due to smoothing

Now we will look at the improvement in the estimate of x_j due to smoothing. The estimate $\hat{x}_j^-$ is the standard *a priori* Kalman filter estimate of x_j , and the estimate $\hat{x}_{j,k+1}$ is the smoothed estimate after measurements up to and including time k have been processed. In other words, $\hat{x}_{j,k}$ uses $(k + 1 - j)$ more measurements to obtain the estimate of x_j than $\hat{x}_j^-$ uses. How much more accurate can we expect our estimate to be with the use of these additional $(k + 1 - j)$ measurements? The estimation accuracy can be measured by the covariance. The improvement in estimation accuracy due to smoothing is equal to the standard estimation covariance P_j minus the smoothed estimation covariance Π_{k+1} . We can use Equations (9.24) and (9.25) to write this improvement as

$$\begin{aligned}
 P_j - \Pi_{k+1} &= \Pi_j - \left(\Pi_j - \sum_{i=j}^k \Sigma_i H_i^T \lambda_i^T \right) \\
 &= \sum_{i=j}^k \Sigma_i H_i^T \lambda_i^T
 \end{aligned} \tag{9.26}$$

Now assume for purposes of additional analysis that the system is time-invariant and the covariance of the standard filter has reached steady state at time j . Then we have

$$\lim_{k \rightarrow \infty} P_k^- = P \tag{9.27}$$

From Equation (9.25) we see that

$$\Sigma_{k+1} = \Sigma_k (F - LH)^T \tag{9.28}$$

where Σ is initialized as $\Sigma_j = P$. Combining this expression for Σ_{k+1} with its initial value, we see that

$$\begin{aligned}
 \Sigma_{k+1} &= P [(F - LH)^T]^{k+1-j} \\
 &= P (\tilde{F}^T)^{k+1-j}
 \end{aligned} \tag{9.29}$$

where $\tilde{F}$ is defined by the above equation. Now substitute this expression, and the expression for λ from Equation (9.25), into Equation (9.26) to obtain

$$\begin{aligned} P_j - \Pi_{k+1} &= \sum_{i=j}^k \Sigma_i H^T \lambda^T \\ &= P \left[\sum_{i=j}^k \left(\tilde{F}^T \right)^{i-j} H^T (H P H^T + R)^{-1} H \tilde{F}^{i-j} \right] P \quad (9.30) \end{aligned}$$

The quantity on the right side of this equation is positive definite, which shows that the smoothed estimate of x_j is always better than the standard Kalman filter estimate. In other words, $(P_j - \Pi_{k+1}) > 0$, which implies that $\Pi_{k+1} < P_j$. Furthermore, the quantity on the right side is a sum of positive definite matrices, which shows that the larger the value of k (i.e., the more measurements that we use to obtain our smoothed estimate), the greater the improvement in the estimation accuracy. Also note from the above that the quantity $(H P H^T + R)$ inside the summation is inverted. This shows that as R increases, the quantity on the right side decreases. In the limit we see from Equation (9.30) that

$$\lim_{R \rightarrow \infty} (P_j - \Pi_{k+1}) = 0 \quad (9.31)$$

This illustrates the general principle that the larger the measurement noise, the smaller the improvement in estimation accuracy that we can obtain by smoothing. This is intuitive because large measurement noise means that additional measurements will not provide much improvement to our estimation accuracy.

■ EXAMPLE 9.1

In this example, we will see the improvement due to smoothing that can be obtained for a vehicle navigation problem. This is a second-order Newtonian system where $x(1)$ is position and $x(2)$ is velocity. The input is comprised of a commanded acceleration u plus acceleration noise $\tilde{u}$. The measurement y is a noisy measurement of position. After discretizing with a step size of T , the system equations can be written as

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} (u_k + \tilde{u}_k) \\ &= \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} u_k + w_k \\ y_k &= \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + v_k \end{aligned} \quad (9.32)$$

Note that the process noise w_k is given as

$$w_k = \begin{bmatrix} T^2/2 \\ T \end{bmatrix} \tilde{u}_k \quad (9.33)$$

Now suppose the acceleration noise $\tilde{u}_k$ has a standard deviation of a . We obtain the process noise covariance as follows:

$$\begin{aligned}
Q_k &= E(w_k w_k^T) \\
&= \begin{bmatrix} T^4/4 & T^3/2 \\ T^3/2 & T^2 \end{bmatrix} E(\tilde{u}_k^2) \\
&= a^2 \begin{bmatrix} T^4/4 & T^3/2 \\ T^3/2 & T^2 \end{bmatrix}
\end{aligned} \tag{9.34}$$

The percent improvement due to smoothing can be defined as

$$\text{Percent Improvement} = \frac{100 \operatorname{Tr}(P_j - \Pi_{k+1})}{\operatorname{Tr}(P_j)} \tag{9.35}$$

where j is the point which is being smoothed, and k is the number of measurements that are processed by the smoother. We can run the fixed-point smoother given by Equation (9.25) in order to smooth the position and velocity estimate at any desired time. Suppose we use the smoother equations to smooth the estimate at the second time step ($k = 1$). If we use measurements at times up to and including 10 seconds to estimate x_1 , then our estimate is denoted as $\hat{x}_{1,101}$. In this case, Table 9.1 shows the percent improvement due to smoothing after 10 seconds when the time step $T = 0.1$ and the acceleration noise standard deviation $a = 0.2$. As expected from the results of the previous subsection, we see that the improvement due to smoothing is more dramatic for small measurement noise.

Table 9.1 Improvement due to smoothing the state at the first time step after 10 seconds for Example 9.1. The improvement due to smoothing is more noticeable when the measurement noise is small.

Measurement noise standard deviation	Percent Improvement
0.1	99.7
1	96.6
10	59.3
100	13.7
1000	0.2

Figure 9.5 shows the trace of Π_k , which is the covariance of the estimation error of the state at the first time step. As time progresses, our estimate of the state at the first time step improves. After 10 seconds of additional measurements, the estimate of the state at the first time step has improved by 96.6% relative to the standard Kalman filter estimate. Figure 9.6 shows the smoothed estimation error of the position and velocity of the first time step. We see that processing more measurements decreases the estimation-error covariance.

In general, the smoothed estimation errors shown in Figure 9.6 will converge to nonzero values. The estimation errors are zero-mean, but not for

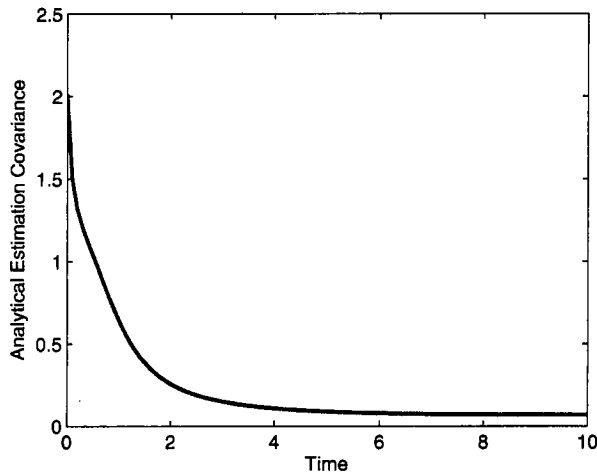


Figure 9.5 This shows the trace of the estimation-error covariance of the smoothed estimate of the state at the first time step for Example 9.1. As time progresses and we process more measurements, the covariance decreases, eventually reaching steady state.

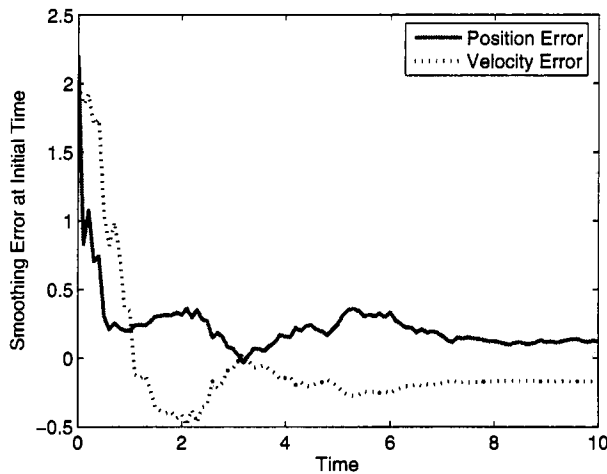


Figure 9.6 This shows typical estimation errors of the smoothed estimate of the state at the first time step for Example 9.1. As time progresses and we process more measurements, the estimation error decreases, and its standard deviation eventually reaches steady state.

any particular simulation. The estimation errors are zero-mean when averaged over many simulations. The system discussed here was simulated 1000 times and the variance of the estimation errors $(x_1 - \hat{x}_{1,101})$ were computed numerically to be equal to 0.054 and 0.012 for the two states. The diagonal elements of Π_{101} were equal to 0.057 and 0.012.

▽▽▽